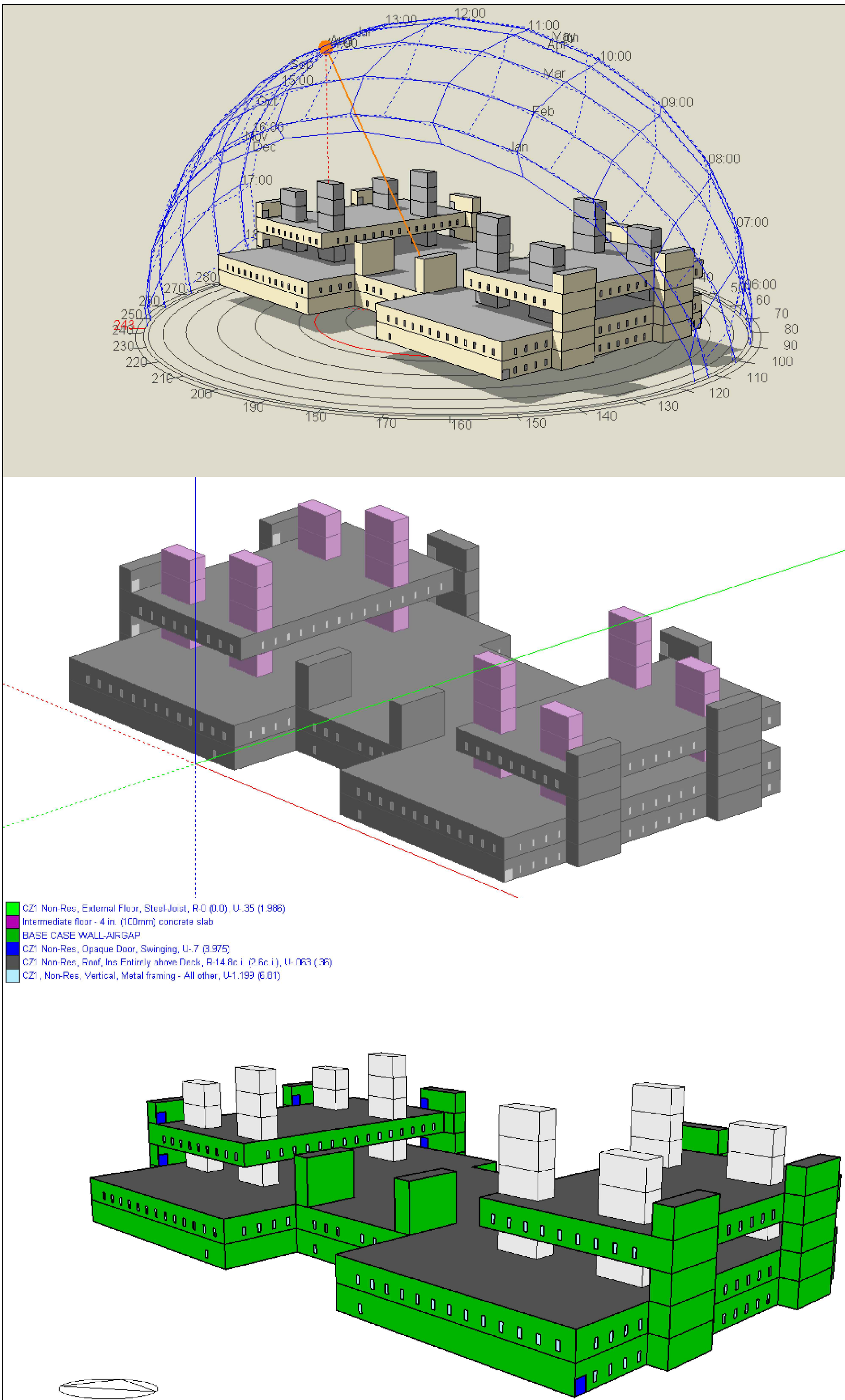




Activity (constant in all cases)

Parameters	Input
Occupancy Density (m ² / person)	15
Metabolic Factor	0.9
Occupancy Schedule	24x7 at 50%
DHW Consumption Rate (l/m ² -day)	0.110
Misc. Power Density (W/m ²)	3
Target Illuminance (lux)	300
Cooling Setpoint (°C)	24
Cooling Setback (°C)	28
Heating Setpoint (°C)	20
Heating Setback (°C)	16
RH Humidification (%)	40
RH Dehumidification (%)	60
Min Fresh Air (l/m ² -person)	5
Mech Vent / Area (l/s-m ²)	0.6

Construction & Openings

Gravel 50mm thick.
Light weight concrete 100mm thick.
Bitumate primer
Reinforced Concrete 240 mm thick.

U-VALUE= 1.324

PLASTER 10mm THICK.
CMU 100mm THICK.
AIR GAP 50mm
CMU 100mm THICK.
PLASTER 10mm THICK.

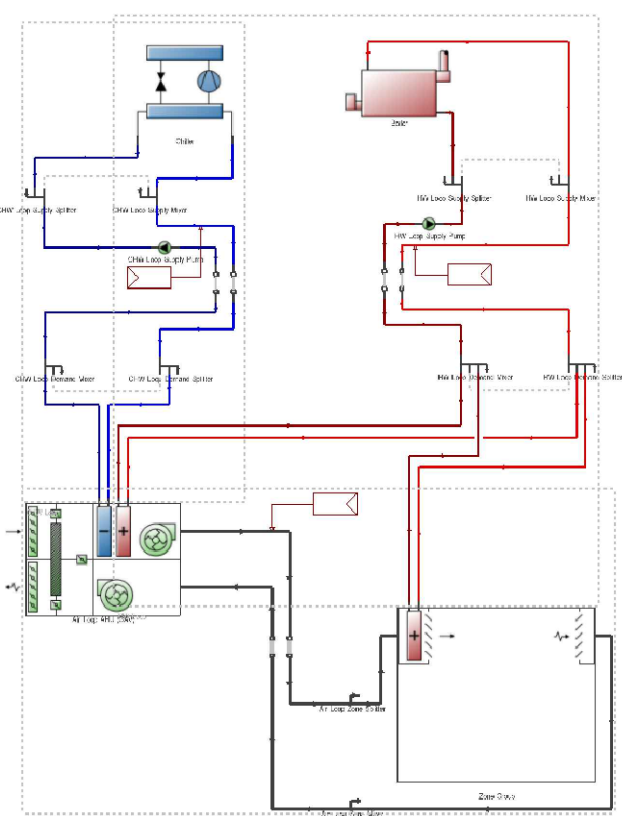
U-VALUE= 1.634

Clear Glass 6mm

U-VALUE= 5.778
SHGC= 0.819

HVAC & Lighting

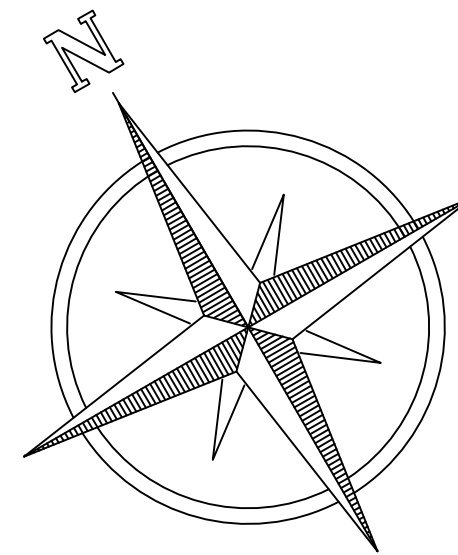
HVAC Parameters	Input
HVAC System	CAV w A/C Chillers
Operation Schedule	24x7 at 50%
Mech Vent. (ac/h)	12
Boiler Nom. thrm, EFF.	0.75
Supp. Air Temp (°C)	35
Chiller CoP	1.5
Supp. Air Temp (°C)	12



Lighting Parameters	Input
Luminaire Type	Halogen
Lighting Density(W/m ²)	35
Luminanre Type	Recessed
Control Type	N/A
Operation Schedule	24x7



COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING

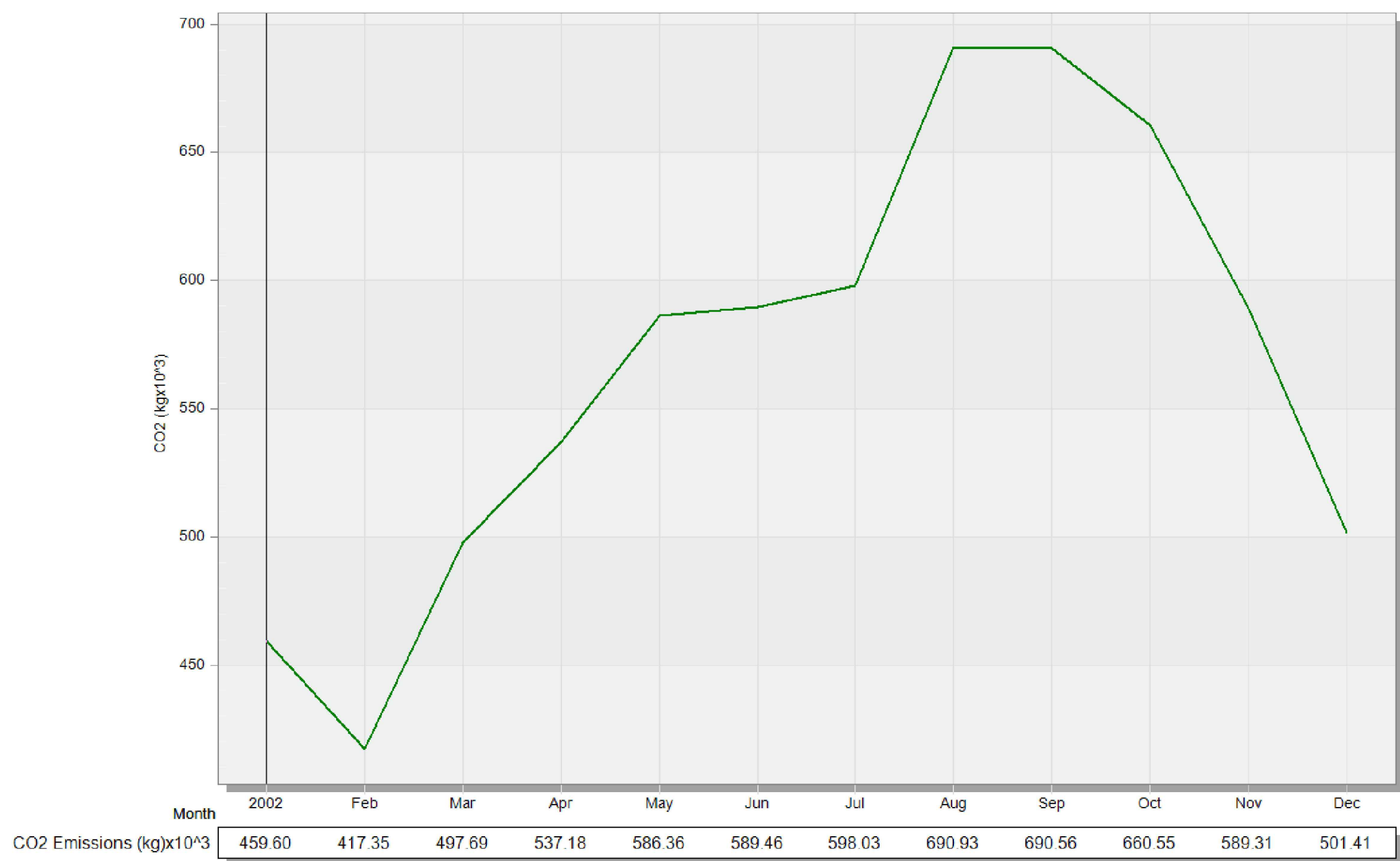


ALL UNITS IN METER

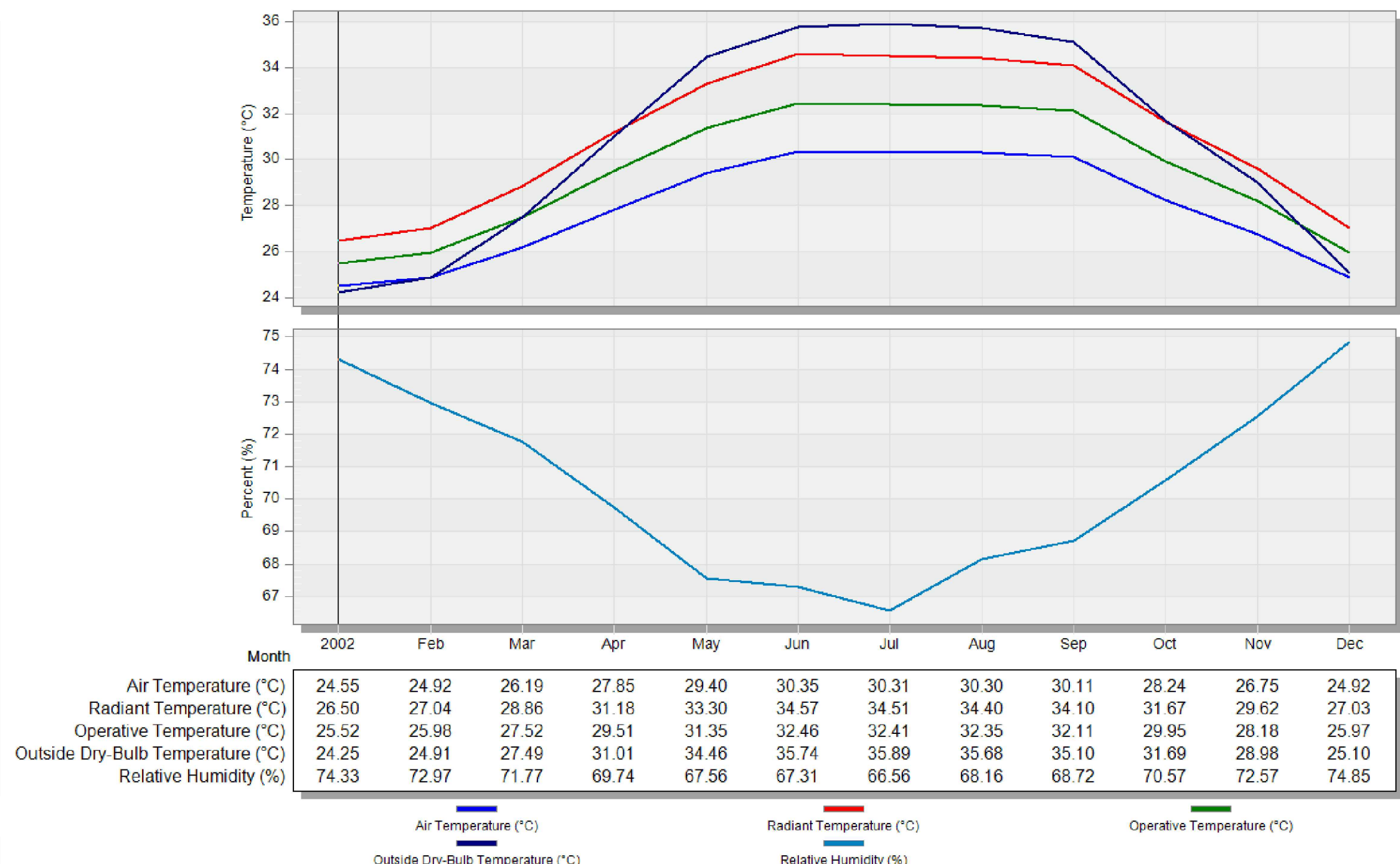
Results of Base Case

Simulation Parameters	Output
Total Site Energy [kWh]	11,251,564.23
Energy Per Total Building Area (kWh/m ²)	634.80
Air Flow / Area (l/s-m ²)	5.27
Design Cooling Load / Area (W/m ²)	114.9
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55
CO ₂ Emissions (Metric Tons)	6,820

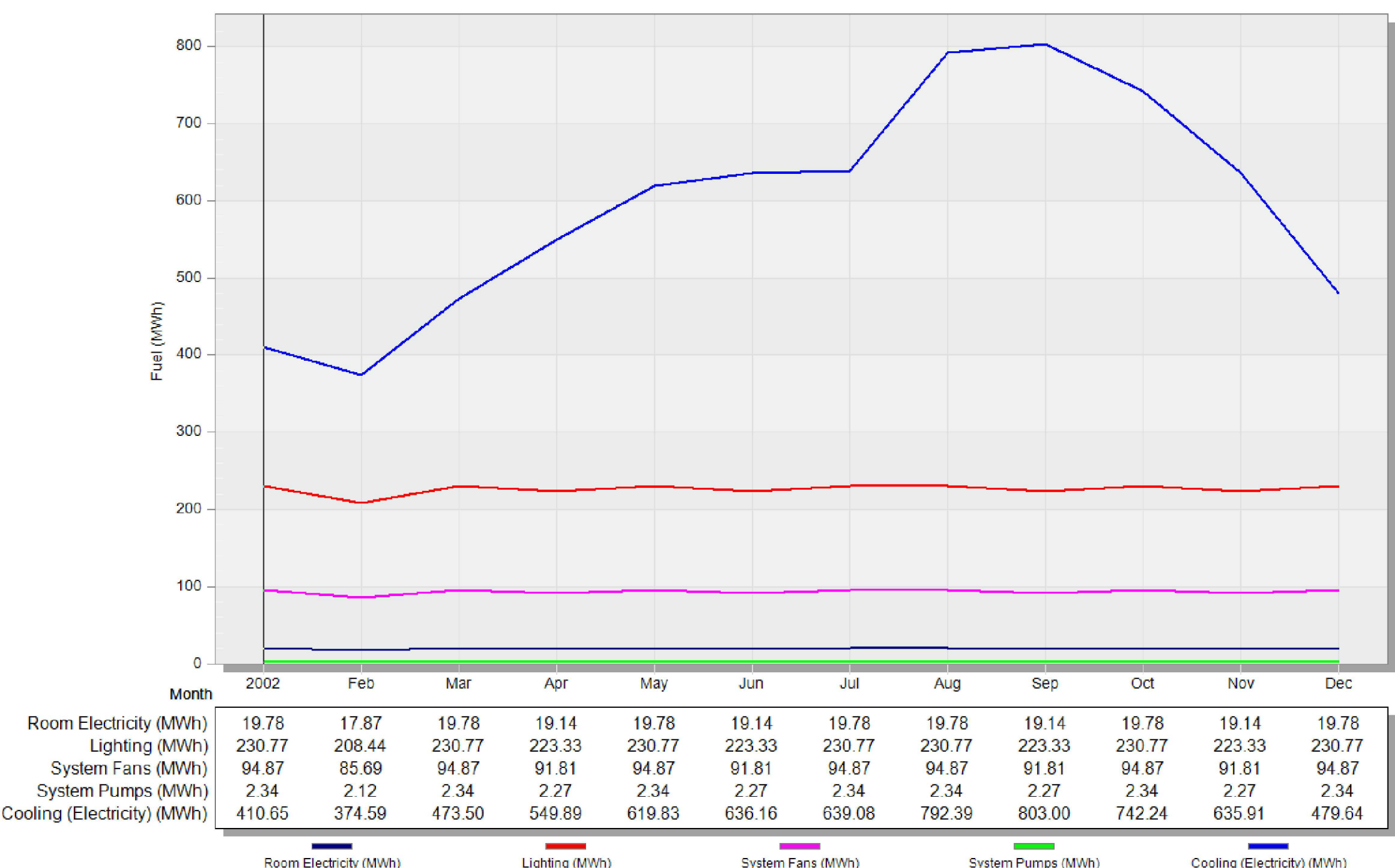
CO₂ Emissions



Comfort



Fuel Breakdown



Design Studio (10), Technical
Project, 2025/2026

INSTRUCTORS:

Prof. Othman Alshamrani (Co)
Prof. Mohammed Essam Shaawat
Dr. M. Abdul Mujeebu
Dr. Mohammad Farooqi
Dr. Saiful Islam
Dr. Abdulaziz Almohassen
Dr. Mohammed Alkhater
Dr. Musaad Al-hawas
Dr. Fahad Alyami
Engr. Meqdad Hasan
Engr. Rehan Jamil

DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Emergency Hospital

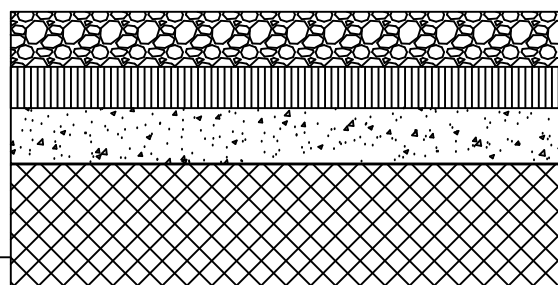
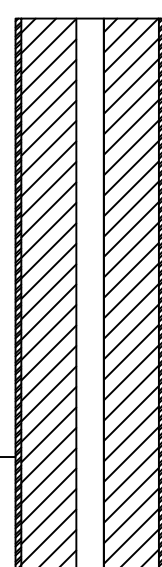
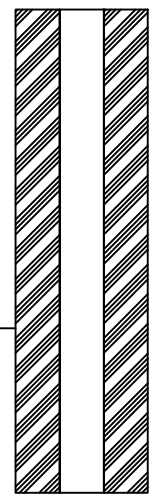
DRAWING NAME:
BASE CASE

SCALE:

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SHEET NUMBER:
A60

Construction & Openings

Opaque Elements	Assembly Max U-Value W/m ² .°C	Insulation Min R-Value m ² .°C/W
Roofs		
Insulation Entirely above Deck	U-0.357	R-2.66 c.i
Walls, Above-Grade		
Mass	U-1.191	R-0.60 c.i.
Gravel 100mm thick. Thermal insulation 50mm thick. Light weight concrete 100mm thick. Bitumate primer Reinforced Concrete 240 mm thick.		
U-VALUE= 0.295		
PLASTER 10mm THICK. CMU 100mm THICK. AIR GAP 50mm CMU 100mm THICK. PLASTER 10mm THICK.		
U-VALUE= 1.174		
Fenestration	Assembly Max U-Value W/m².°C	Assembly Max. SHGC
Vertical Glazing, 0% to 40% of Wall		
Nonmetal framing (all)	U-2.89	
Metal framing (curtainwall/storefront)d	U-2.89	0.25
Low-E coated Glass 6mm Air Gap 6mm Clear Glass 6mm		
U-VALUE= 2.342 SHGC= 0.248		

Simulation Parameters	Output Before	Output After
Total Site Energy [kWh]	11,251,564.23	10,655,180.61 (-5.30%)
Energy Per Total Building Area (kWh/m ²)	634.80	601.15 (-5.30%)
Air Flow / Area (l/s-m ²)	5.27	4.99 (-5.31%)
Design Cooling Load / Area (W/m ²)	114.9	109.1 (-5.05%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,409,657.8 (-190,842.75)
CO ₂ Emissions (Metric Tons)	6,820	6,460 (-5.28%)

<u>Simulation Parameters</u>	<u>Output Before</u>	<u>Output After</u>
Total Site Energy [kWh]	11,251,564.23	11,102,946.31 (-1.32%)
Energy Per Total Building Area (kWh/m ²)	634.80	626.41 (-1.32%)
Air Flow / Area (l/s-m ²)	5.27	5.18 (-1.71%)
Design Cooling Load / Area (W/m ²)	114.9	113.2 (-1.48%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,552,942.82 (-\$47,557.73)
CO ₂ Emissions (Metric Tons)	6,820	6,730 (-1.32%)

<u>Simulation Parameters</u>	<u>Output Before</u>	<u>Output After</u>
Total Site Energy [kWh]	11,251,564.23	11,204,247.08 (-0.42%)
Energy Per Total Building Area (kWh/m ²)	634.80	632.13 (-0.42%)
Air Flow / Area (l/s-m ²)	5.27	5.23 (-0.76%)
Design Cooling Load / Area (W/m ²)	114.9	114.2 (-0.61%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,585,359.07 (-\$15,141.48)
CO ₂ Emissions (Metric Tons)	6,820	6,790 (-0.44%)

<u>Simulation Parameters</u>	<u>Output Before</u>	<u>Output After</u>
Total Site Energy [kWh]	11,251,564.23	10,419,712.51 (-7.39%)
Energy Per Total Building Area (kWh/m ²)	634.80	587.87 (-7.39%)
Air Flow / Area (l/s-m ²)	5.27	4.85 (-7.97%)
Design Cooling Load / Area (W/m ²)	114.9	105.4 (-8.27%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,334,308.00 (-266,192.55)
CO ₂ Emissions (Metric Tons)	6,820	6,310 (-7.48%)

	HVAC & Lighting
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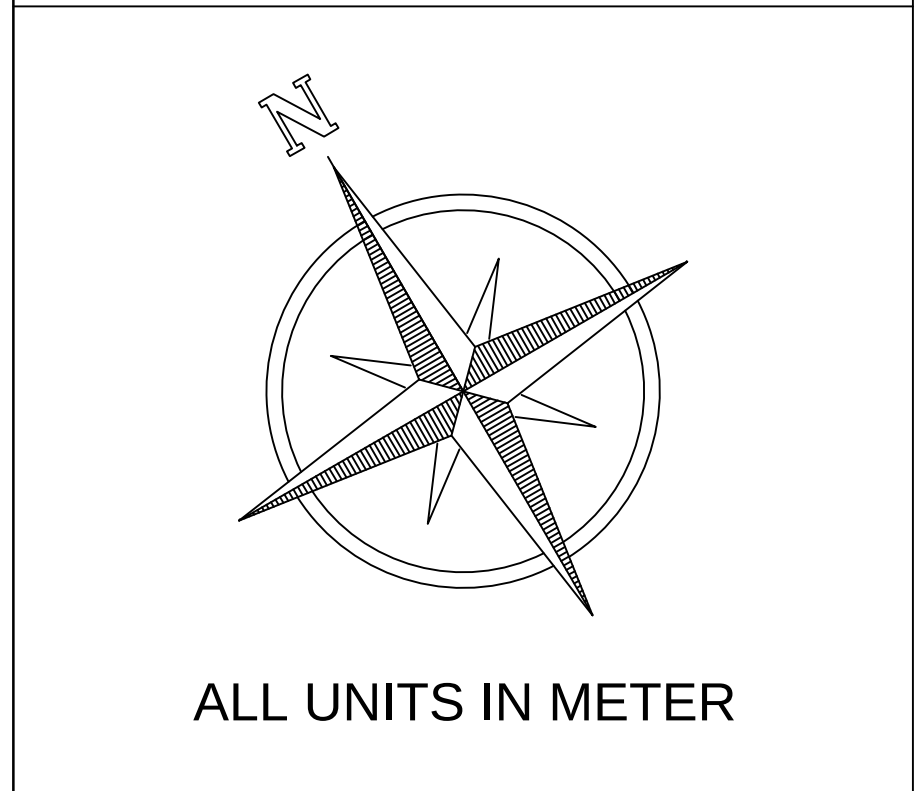
<u>HVAC Parameters</u>	<u>Input</u>
HVAC System	CAV w A/C Chillers
Operation Schedule	24x7
Mech Vent. (ac/h)	12
Boiler Nom. thrm, EFF.	0.81
Supp. Air Temp (°C)	35
Chiller CoP	2.1
Supp. Air Temp (°C)	12

<u>Lighting Parameters</u>	<u>Input</u>
Luminare Type	LED
Lighting Density(W/m ²)	25
Luminanre Type	Recessed
Control Type	N/A
Operation Schedule	24x7

SBC HVAC Sub-Case		
Simulation Parameters	Output Before	Output After
Total Site Energy [kWh]	11,251,564.23	9,206,740.20 (-18.17%)
Energy Per Total Building Area (kWh/m ²)	634.80	519.43 (-18.17%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	2,946,156.89 (-654,343.69)
CO ₂ Emissions (Metric Tons)	6,820	5,580 (-18.18%)

<u>Simulation Parameters</u>	<u>Output Before</u>	<u>Output After</u>
Total Site Energy [kWh]	11,251,564.23	9,989,863.08 (-11.21%)
Energy Per Total Building Area (kWh/m ²)	634.80	563.62(-11.21%)
Air Flow / Area (l/s-m ²)	5.27	5.01 (-4.93%)
Design Cooling Load / Area (W/m ²)	114.9	109.8 (-4.44%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,196,756.19(-403,744.37)
CO ₂ Emissions (Metric Tons)	6,820	6,050 (-11.29%)

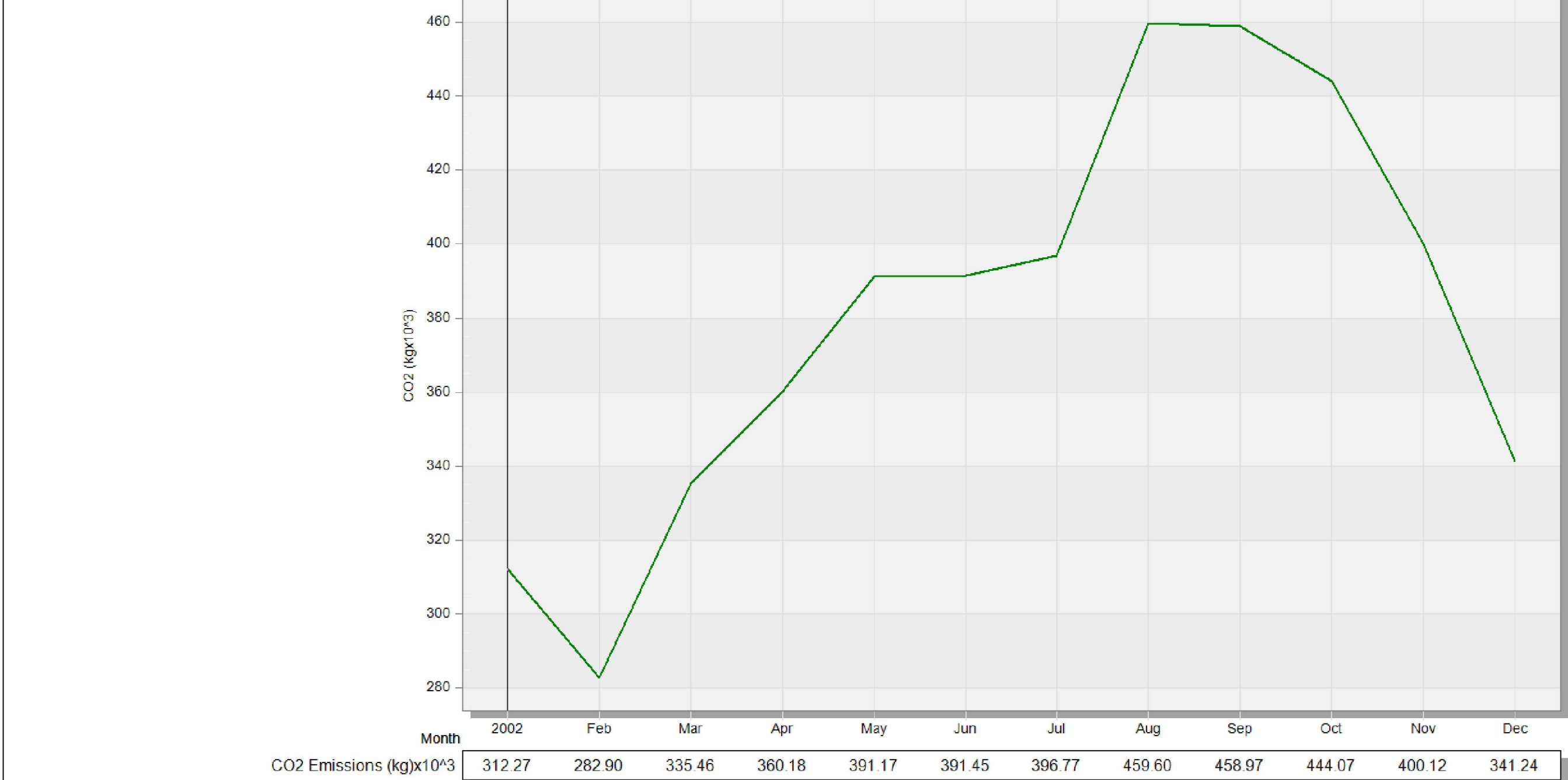
SBC Active-Case		
<u>Simulation Parameters</u>	<u>Output Before</u>	<u>Output After</u>
Total Site Energy [kWh]	11,251,564.23	8,055,844.56 (-28.40%)
Energy Per Total Building Area (kWh/m ²)	634.80	454.50(-28.40%)
Air Flow / Area (l/s-m ²)	5.27	5.01 (-4.93%)
Design Cooling Load / Area (W/m ²)	114.9	109.8 (-4.44%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	2,577,870.26 (-1,022,630.29)
CO ₂ Emissions (Metric Tons)	6,820	4,880 (-3.68%)



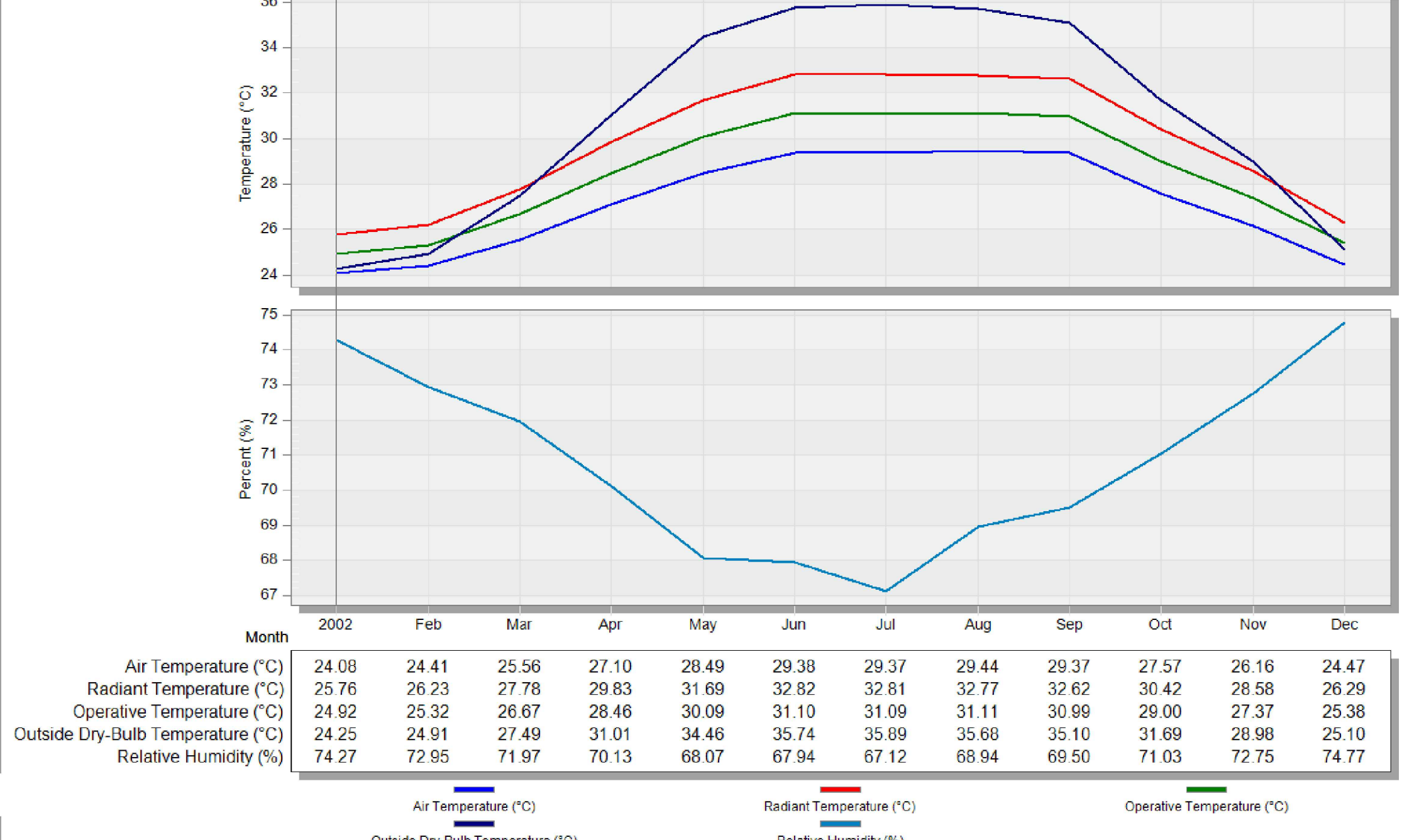
Results of SBC Case	
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Simulation Parameters	Output Before	Output After
Total Site Energy [kWh]	11,251,564.23	7,548,217.84 (-32.91%)
Energy Per Total Building Area (kWh/m²)	634.80	425.86 (-32.91%)
Air Flow / Area (l/s-m²)	5.27	4.58 (-13.09%)
Design Cooling Load / Area (W/m²)	114.9	96.5 (-16.01%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	2,415,429.71 (-1,185,070.84)
CO2 Emissions (Metric Tons)	6,820	4,570 (-32.99%)

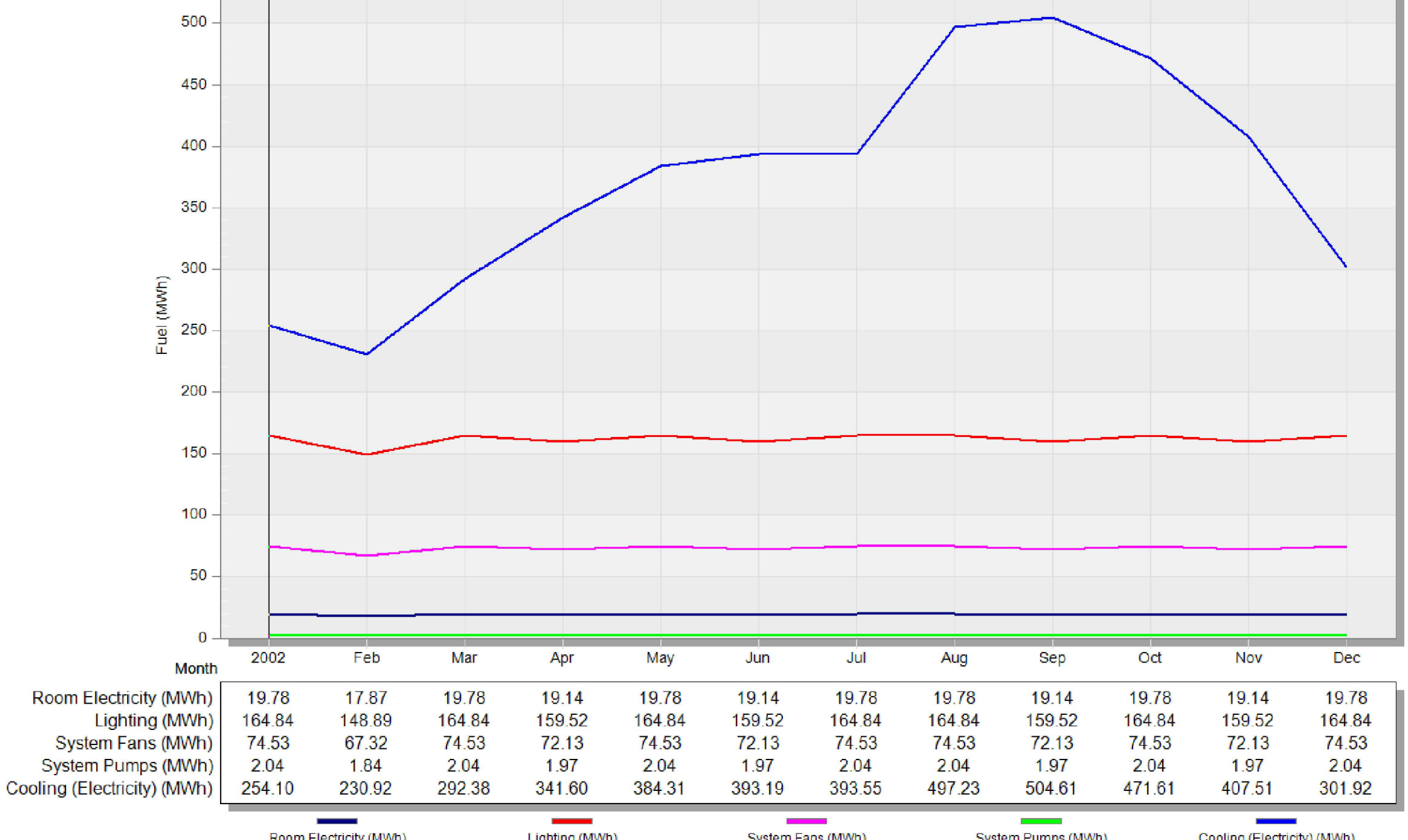
CO₂ Emissions



Comfort



Fuel Breakdown



Design Studio (10), Technical Project, 2025/2026
--

INSTRUCTORS:

Prof. Othman Alshamrani (Co)
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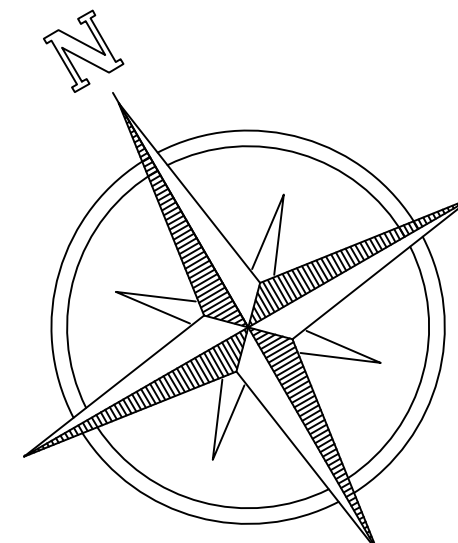
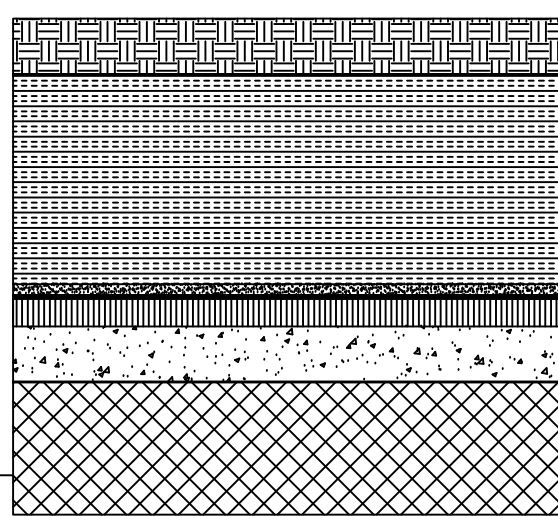
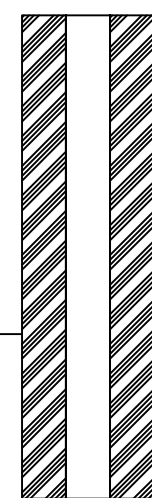
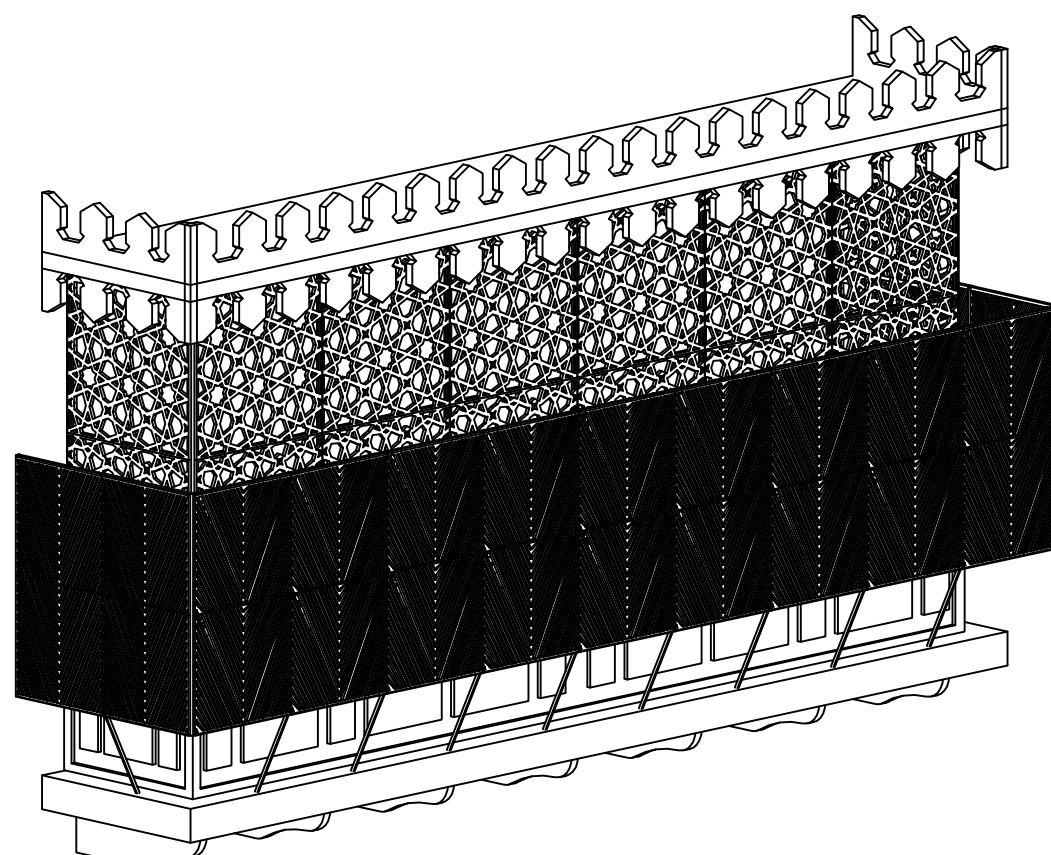
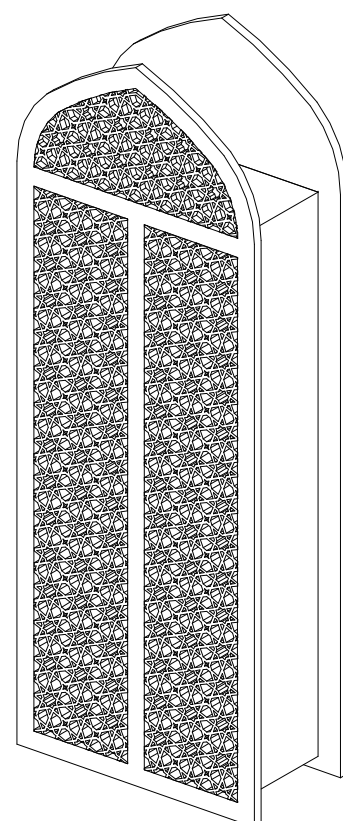
DONE BY:
ABDULELAH ALQABBANI


PROJECT:
Emergency Hospital

DRAWING NAME:
SBC CASE

SCALE: --

SHEET NUMBER:
A61

CASES	CHANGE	RESULTS	 جامعة الإمام عبد الرحمن بن فيصل IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY COLLEGE OF ARCH. & PLAN DEPT. OF BUILDING ENGINEERING  ALL UNITS IN METER																																								
Roof Sub-Case	SOIL 100mm thick. Bitumate Sheeting GEO-FOAM FILLER 98%AIR 380mm thick. SAND CEMENT SECREED 20mm Seperation Layer (polyethylene 150 mlc.) Water proof membrane 5mm thick. Expanded PolyUrethane 50mm thick. Aerated Lightweight concrete 100mm thick. Bitumate primer Reinforced Concrete 220 mm thick.  U-VALUE= 0.070	Best Roof Sub-Case <table><tr><th>Simulation Parameters</th><th>Output Base</th><th>Output SBC</th><th>Output Best</th></tr><tr><td>Total Site Energy [kWh]</td><td>11,251,564.23</td><td>10,655,180.61 (-5.30%)</td><td>10,501,848.65 (-6.66%)</td></tr><tr><td>Energy Per Total Building Area (kWh/m²)</td><td>634.80</td><td>601.15 (-5.30%)</td><td>592.50 (-6.66%)</td></tr><tr><td>Air Flow / Area (l/s-m²)</td><td>5.27</td><td>4.99 (-5.31%)</td><td>4.89 (-7.21%)</td></tr><tr><td>Design Cooling Load / Area (W/m²)</td><td>114.9</td><td>109.1 (-5.05%)</td><td>105.7 (-8.01%)</td></tr><tr><td>Annual Tariff for Gov. Bldgs. (SAR/Year)</td><td>3,600,500.55</td><td>3,409,657.8 (-190,842.75)</td><td>3,360,591.57 (-239,908.98)</td></tr><tr><td>CO₂ Emissions (Metric Tons)</td><td>6,820</td><td>6,460 (-5.28%)</td><td>6,360 (-6.74%)</td></tr></table>		Simulation Parameters	Output Base	Output SBC	Output Best	Total Site Energy [kWh]	11,251,564.23	10,655,180.61 (-5.30%)	10,501,848.65 (-6.66%)	Energy Per Total Building Area (kWh/m ²)	634.80	601.15 (-5.30%)	592.50 (-6.66%)	Air Flow / Area (l/s-m ²)	5.27	4.99 (-5.31%)	4.89 (-7.21%)	Design Cooling Load / Area (W/m ²)	114.9	109.1 (-5.05%)	105.7 (-8.01%)	Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,409,657.8 (-190,842.75)	3,360,591.57 (-239,908.98)	CO ₂ Emissions (Metric Tons)	6,820	6,460 (-5.28%)	6,360 (-6.74%)												
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Wall Sub-Case	TRAVETRINE TILES 10mm THICK. POLYMER-MODIFIEED MORTAR 5mm THICK. CMU 100mm THICK. EXTRUDED POLYSTERINE BOARD 70mm THICK. CMU 100mm THICK. PLASTER 10mm THICK. PAINT 5mm THICK. U-VALUE= 0.263	Best Wall Sub-Case <table><tr><th>Simulation Parameters</th><th>Output Base</th><th>Output SBC</th><th>Output Best</th></tr><tr><td>Total Site Energy [kWh]</td><td>11,251,564.23</td><td>11,102,946.31 (-1.32%)</td><td>10,890,545.68 (-3.21%)</td></tr><tr><td>Energy Per Total Building Area (kWh/m²)</td><td>634.80</td><td>626.41 (-1.32%)</td><td>614.43 (-3.21%)</td></tr><tr><td>Air Flow / Area (l/s-m²)</td><td>5.27</td><td>5.18 (-1.71%)</td><td>5.07 (-3.80%)</td></tr><tr><td>Design Cooling Load / Area (W/m²)</td><td>114.9</td><td>113.2 (-1.48%)</td><td>111.0 (-3.39%)</td></tr><tr><td>Annual Tariff for Gov. Bldgs. (SAR/Year)</td><td>3,600,500.55</td><td>3,552,942.82 (-\$47,557.73)</td><td>3,484,974.62 (-\$115,525.93)</td></tr><tr><td>CO₂ Emissions (Metric Tons)</td><td>6,820</td><td>6,730 (-1.32%)</td><td>6,600 (-3.23%)</td></tr></table>	Simulation Parameters	Output Base	Output SBC	Output Best	Total Site Energy [kWh]	11,251,564.23	11,102,946.31 (-1.32%)	10,890,545.68 (-3.21%)	Energy Per Total Building Area (kWh/m ²)	634.80	626.41 (-1.32%)	614.43 (-3.21%)	Air Flow / Area (l/s-m ²)	5.27	5.18 (-1.71%)	5.07 (-3.80%)	Design Cooling Load / Area (W/m ²)	114.9	113.2 (-1.48%)	111.0 (-3.39%)	Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,552,942.82 (-\$47,557.73)	3,484,974.62 (-\$115,525.93)	CO ₂ Emissions (Metric Tons)	6,820	6,730 (-1.32%)	6,600 (-3.23%)													
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Glazing Sub-Case	Low-E Double coated Glass 6mm Xenon 6mm Low-E Double coated Glass 6mm U-VALUE= 0.831 SHGC= 0.164 	Best Glazing Sub-Case <table><tr><th>Simulation Parameters</th><th>Output Base</th><th>Output SBC</th><th>Output Best</th></tr><tr><td>Total Site Energy [kWh]</td><td>11,251,564.23</td><td>11,204,247.08 (-0.42%)</td><td>11,195,167.17 (-0.50%)</td></tr><tr><td>Energy Per Total Building Area (kWh/m²)</td><td>634.80</td><td>632.13 (-0.42%)</td><td>631.62 (-0.50%)</td></tr><tr><td>Air Flow / Area (l/s-m²)</td><td>5.27</td><td>5.23 (-0.76%)</td><td>5.23 (-0.76%)</td></tr><tr><td>Design Cooling Load / Area (W/m²)</td><td>114.9</td><td>114.2 (-0.61%)</td><td>114.0 (-0.78%)</td></tr><tr><td>Annual Tariff for Gov. Bldgs. (SAR/Year)</td><td>3,600,500.55</td><td>3,585,359.07 (-\$15,141.48)</td><td>3,582,453.5 (-\$18,047.05)</td></tr><tr><td>CO₂ Emissions (Metric Tons)</td><td>6,820</td><td>6,790 (-0.44%)</td><td>6,780 (-0.59%)</td></tr></table>	Simulation Parameters	Output Base	Output SBC	Output Best	Total Site Energy [kWh]	11,251,564.23	11,204,247.08 (-0.42%)	11,195,167.17 (-0.50%)	Energy Per Total Building Area (kWh/m ²)	634.80	632.13 (-0.42%)	631.62 (-0.50%)	Air Flow / Area (l/s-m ²)	5.27	5.23 (-0.76%)	5.23 (-0.76%)	Design Cooling Load / Area (W/m ²)	114.9	114.2 (-0.61%)	114.0 (-0.78%)	Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,585,359.07 (-\$15,141.48)	3,582,453.5 (-\$18,047.05)	CO ₂ Emissions (Metric Tons)	6,820	6,790 (-0.44%)	6,780 (-0.59%)													
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Shading Sub-Case		Best Shading Sub-Case <table><tr><th>Simulation Parameters</th><th>Output Base</th><th>Output Shading only</th><th>Output Best Glazing &shading</th></tr><tr><td>Total Site Energy [kWh]</td><td>11,251,564.23</td><td>11,189,817.55 (-0.55%)</td><td>11,171,358.97 (-0.71%)</td></tr><tr><td>Energy Per Total Building Area (kWh/m²)</td><td>634.80</td><td>631.32 (-0.55%)</td><td>630.27 (-0.71%)</td></tr><tr><td>Air Flow / Area (l/s-m²)</td><td>5.27</td><td>5.23 (-0.76%)</td><td>5.21 (-1.14%)</td></tr><tr><td>Design Cooling Load / Area (W/m²)</td><td>114.9</td><td>114.1 (-0.70%)</td><td>113.7 (-1.04%)</td></tr><tr><td>Annual Tariff for Gov. Bldgs. (SAR/Year)</td><td>3,600,500.55</td><td>3,580,741.62 (-\$19,758.93)</td><td>3,574,834.87 (-\$25,665.68)</td></tr><tr><td>CO₂ Emissions (Metric Tons)</td><td>6,820</td><td>6,780 (-0.59%)</td><td>6,770 (-0.73%)</td></tr></table>	Simulation Parameters	Output Base	Output Shading only	Output Best Glazing &shading	Total Site Energy [kWh]	11,251,564.23	11,189,817.55 (-0.55%)	11,171,358.97 (-0.71%)	Energy Per Total Building Area (kWh/m ²)	634.80	631.32 (-0.55%)	630.27 (-0.71%)	Air Flow / Area (l/s-m ²)	5.27	5.23 (-0.76%)	5.21 (-1.14%)	Design Cooling Load / Area (W/m ²)	114.9	114.1 (-0.70%)	113.7 (-1.04%)	Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,580,741.62 (-\$19,758.93)	3,574,834.87 (-\$25,665.68)	CO ₂ Emissions (Metric Tons)	6,820	6,780 (-0.59%)	6,770 (-0.73%)	Design Studio (10), Technical Project, 2025/2026 INSTRUCTORS: Prof. Othman Alshamrani (Co) Prof. Mohammed Essam Shaawat Dr. M. Abdul Mujeebu Dr. Mohammad Farooqi Dr. Saiful Islam Dr. Abdulaziz Almohassen Dr. Mohammed Alkhater Dr. Musaad Al-hawas Dr. Fahad Alyami Engr. Meqdad Hasan Engr. Rehan Jamil												
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HVAC Parameters	Input																																										
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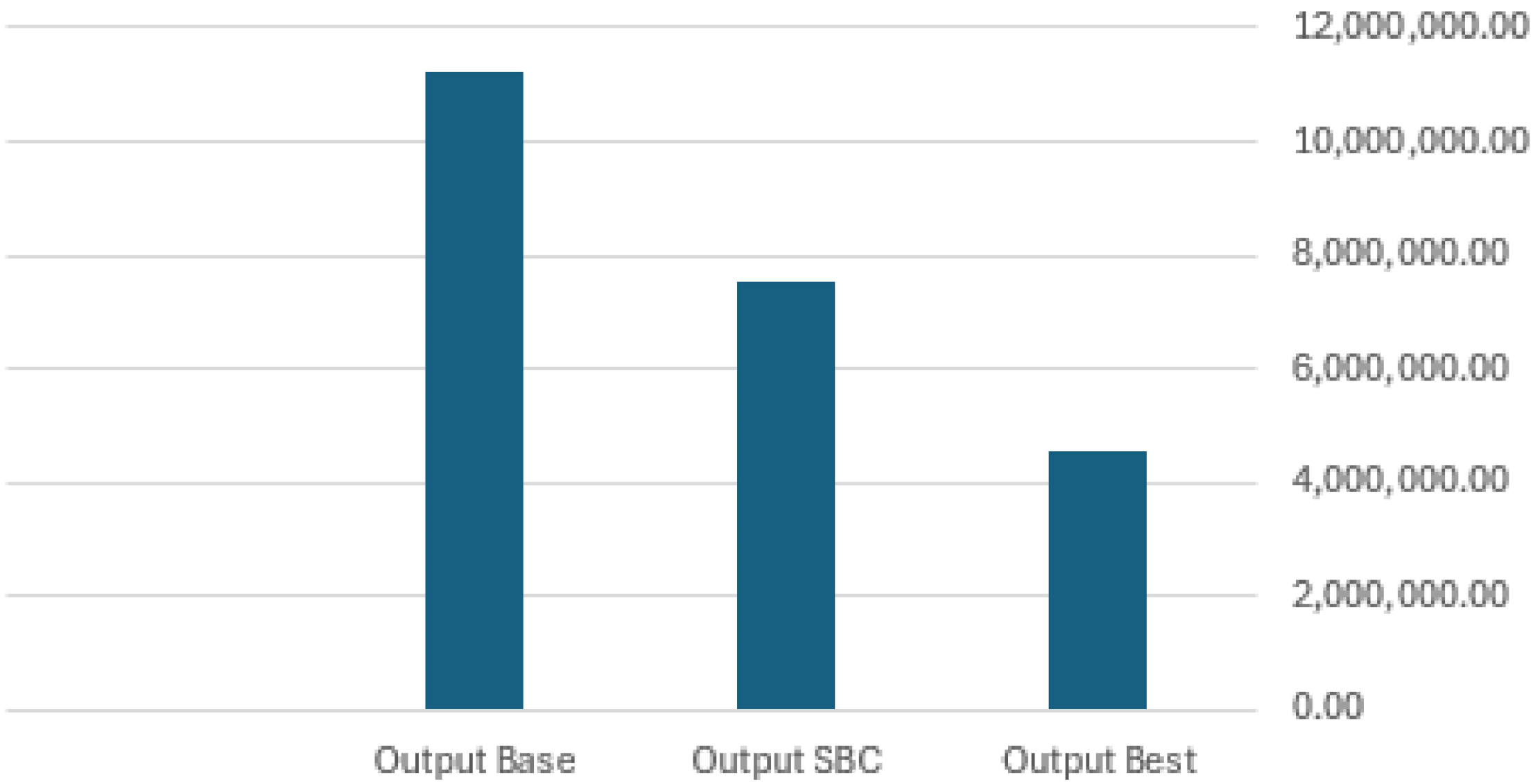
Simulation Parameters	Output Base	Output SBC	Output Best
Total Site Energy [kWh]	11,251,564.23	10,419,712.51 (-7.39%)	10,092,015.57 (-10.31%)
Energy Per Total Building Area (kWh/m²)	634.80	587.87 (-7.39%)	569.38 (-10.31%)
Air Flow / Area (l/s-m²)	5.27	4.85 (-7.97%)	4.60 (-12.71%)
Design Cooling Load / Area (W/m²)	114.9	105.4 (-8.27%)	97.2 (-15.40%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	3,334,308.00 (-266,192.55)	3,229,444.98 (-371,055.57)
CO₂ Emissions (Metric Tons)	6,820	6,310 (-7.48%)	6,120 (-10.26%)

Simulation Parameters	Output Base	Output SBC	Output Best
Total Site Energy [kWh]	11,251,564.23	8,055,844.56 (-28.40%)	4,644,717.33 (-58.72%)
Energy Per Total Building Area (kWh/m²)	634.80	454.50(-28.40%)	262.05(-58.72%)
Air Flow / Area (l/s-m²)	5.27	5.01 (-4.93%)	4.69 (-11.01%)
Design Cooling Load / Area (W/m²)	114.9	109.8 (-4.44%)	101.0 (-12.10%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	2,577,870.26 (-1,022,630.29)	1,486,309.55 (-2,114,191.00)
CO₂ Emissions (Metric Tons)	6,820	4,880 (-3.68%)	2,810 (-58.80%)

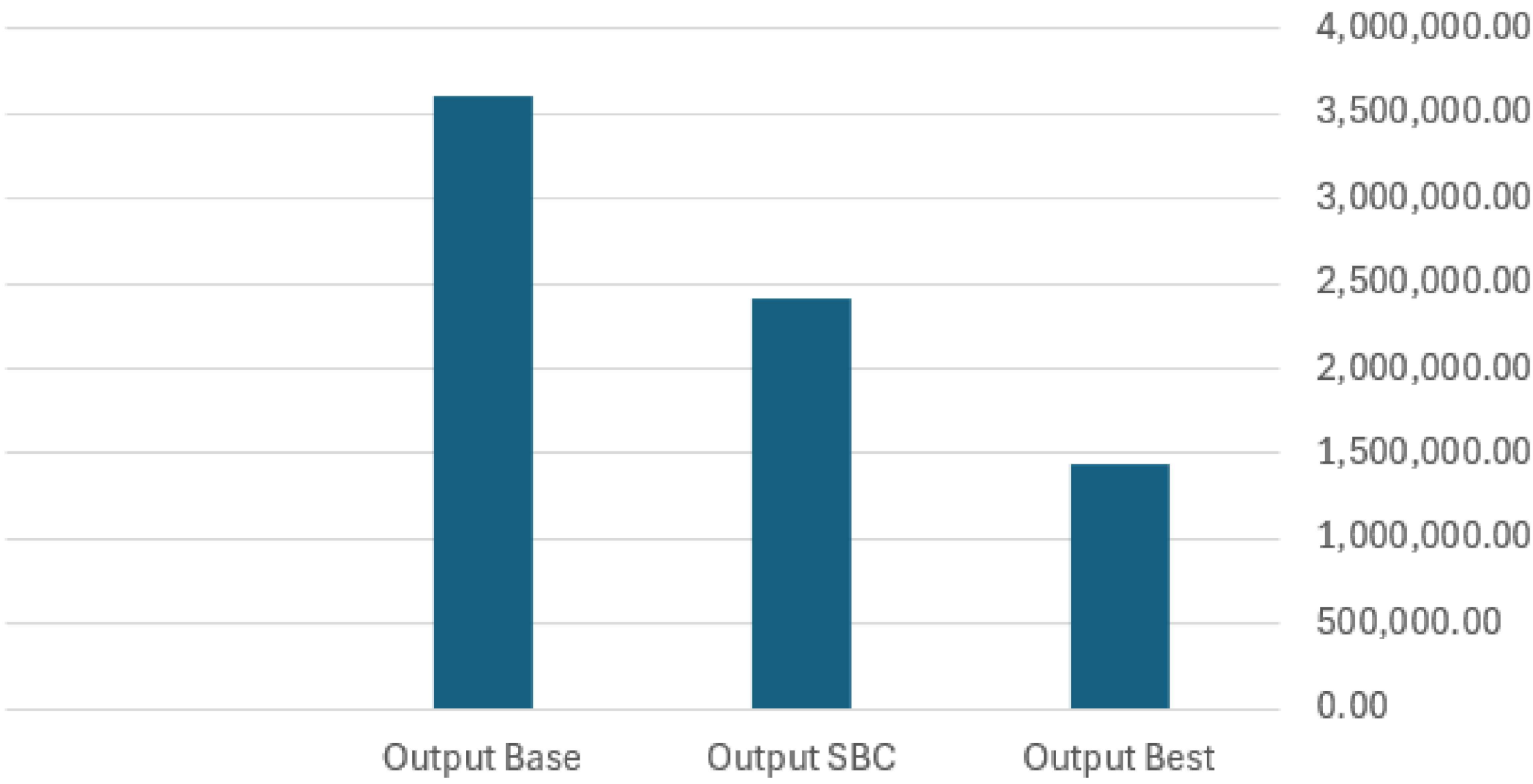
Results of Best Case

Simulation Parameters	Output Base	Output SBC	Output Best
Total Site Energy [kWh]	11,251,564.23	7,548,217.84 (-32.91%)	4,515,270.26 (-59.87%)
Energy Per Total Building Area (kWh/m²)	634.80	425.86 (-32.91%)	254.75 (-59.87%)
Air Flow / Area (l/s-m²)	5.27	4.58 (-13.09%)	4.04 (-23.34%)
Design Cooling Load / Area (W/m²)	114.9	96.5 (-16.01%)	85.1 (-25.94%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	2,415,429.71 (-1,185,070.84)	1,444,886.48 (-2,155,614.07)
CO₂ Emissions (Metric Tons)	6,820	4,570 (-32.99%)	2,740 (-59.82%)

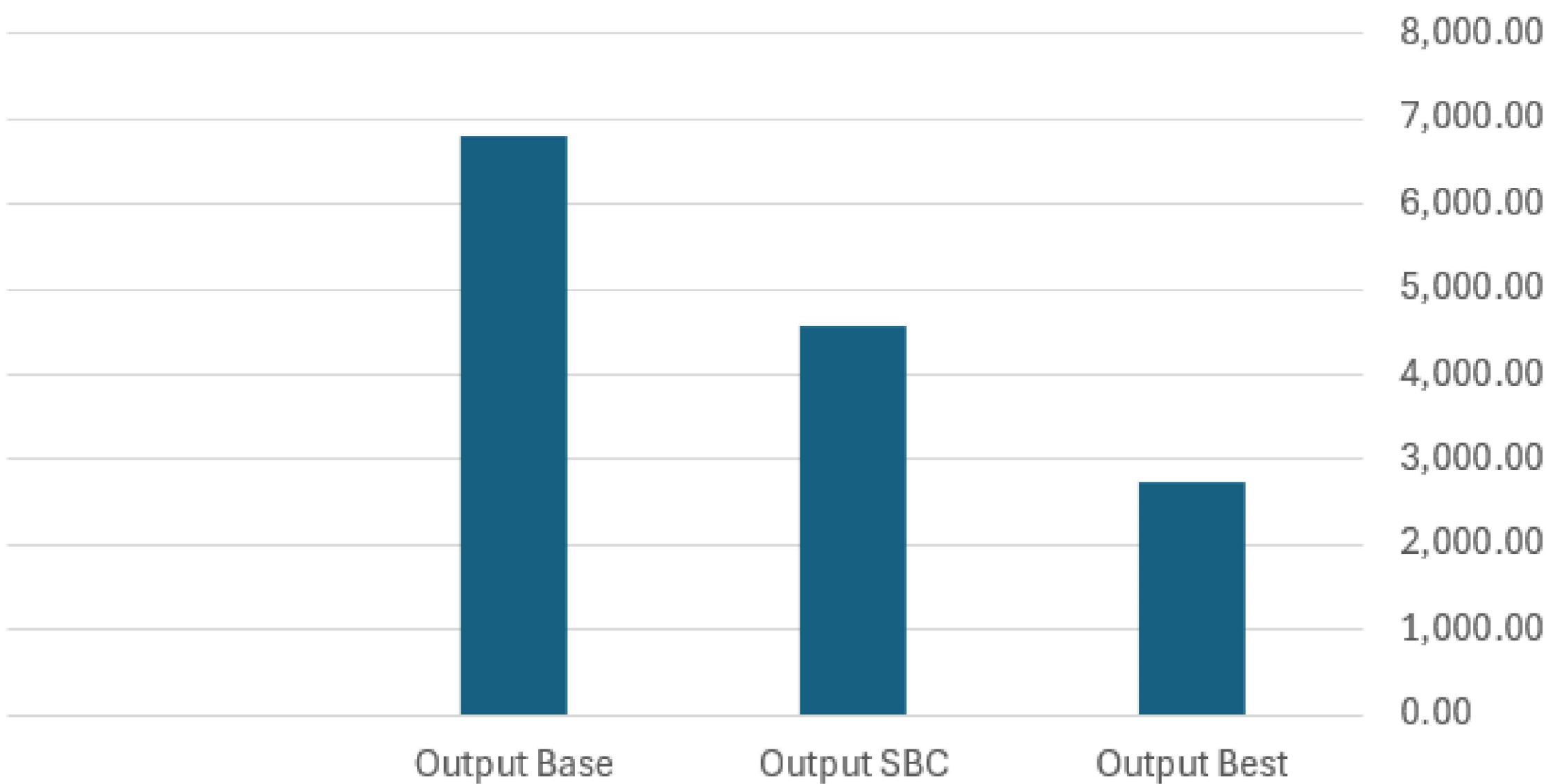
Total Site Energy [kWh]



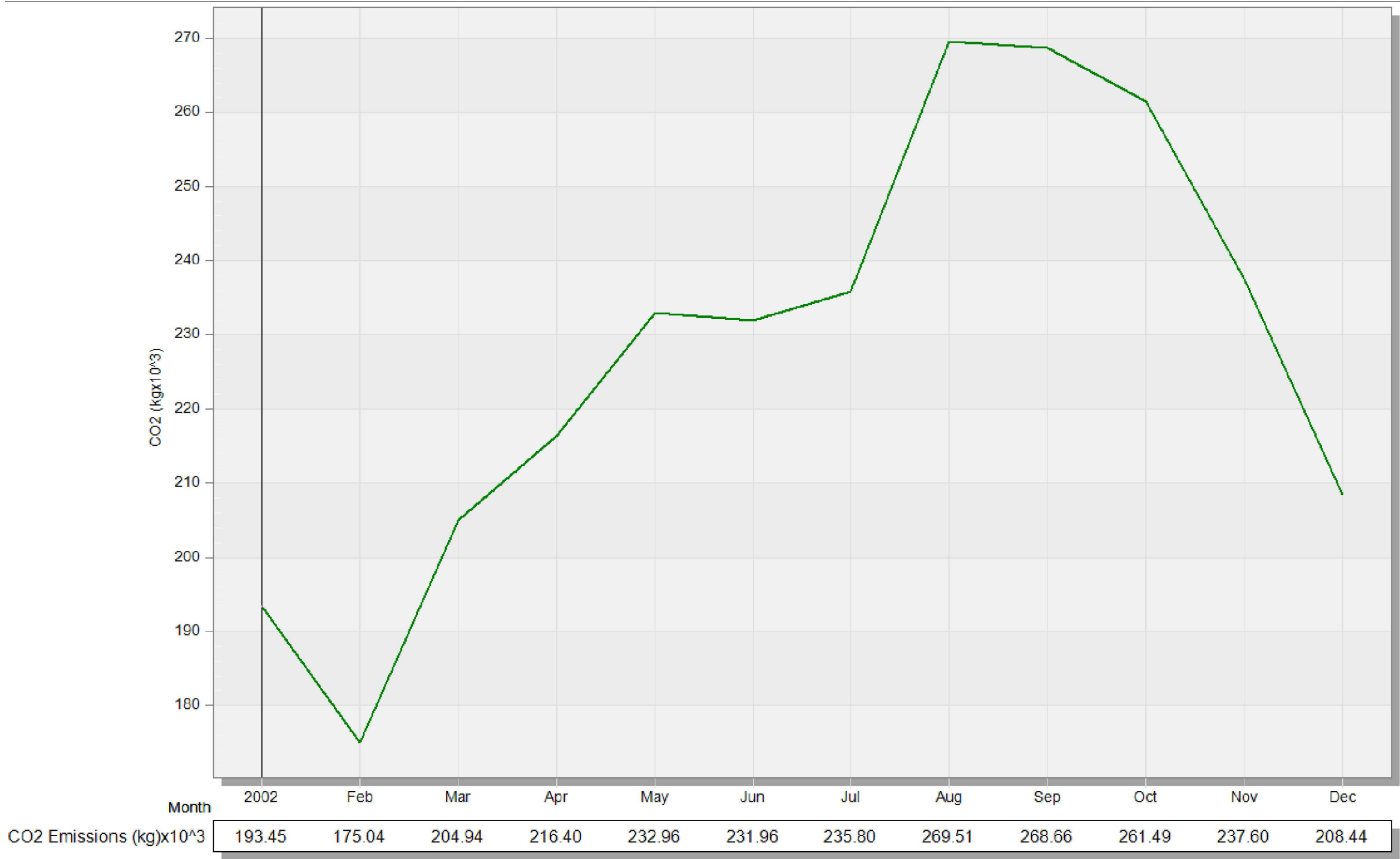
Annual Tariff for Gov. Bldgs. (SAR/Year)



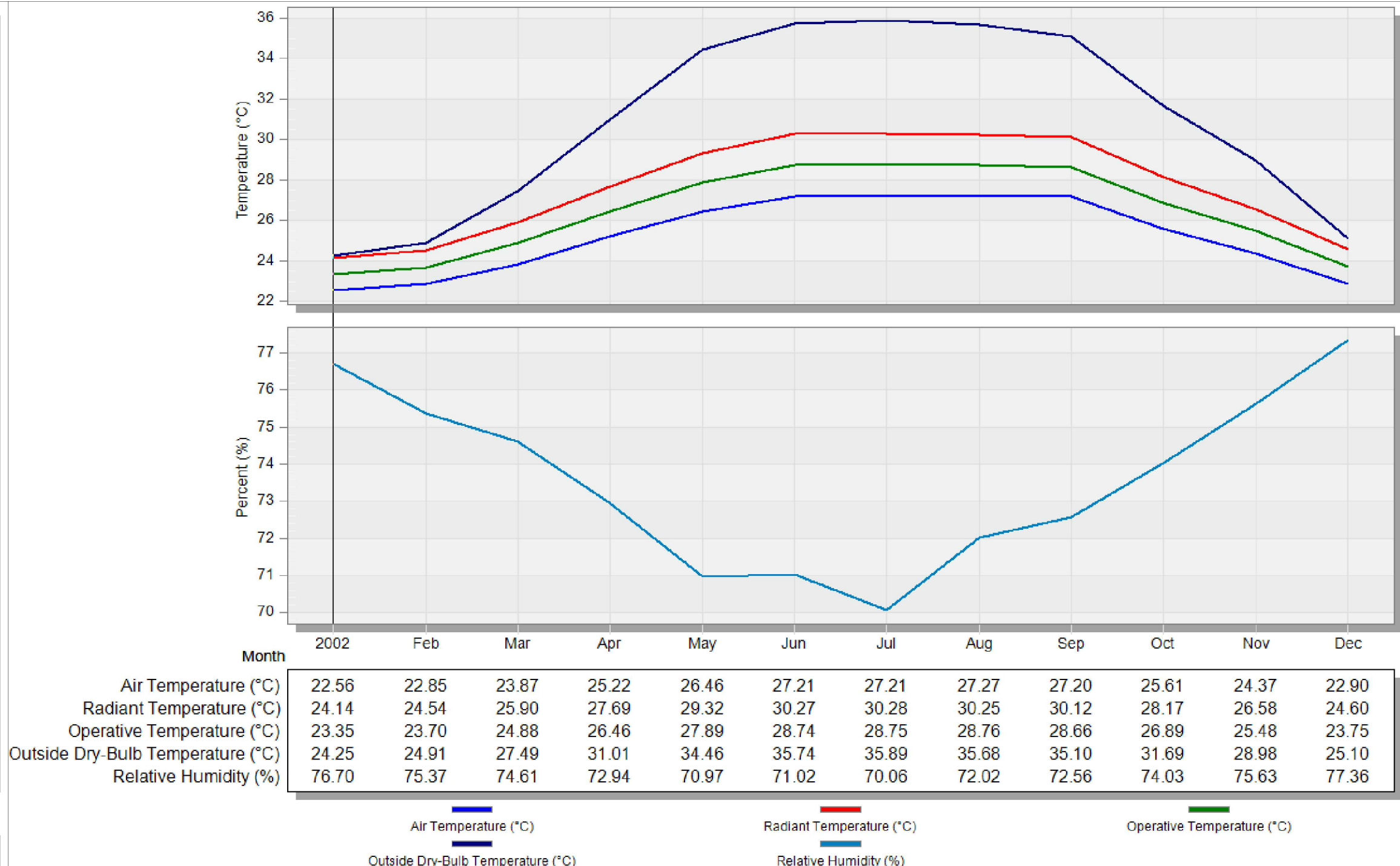
CO2 Emissions (Metric Tons)



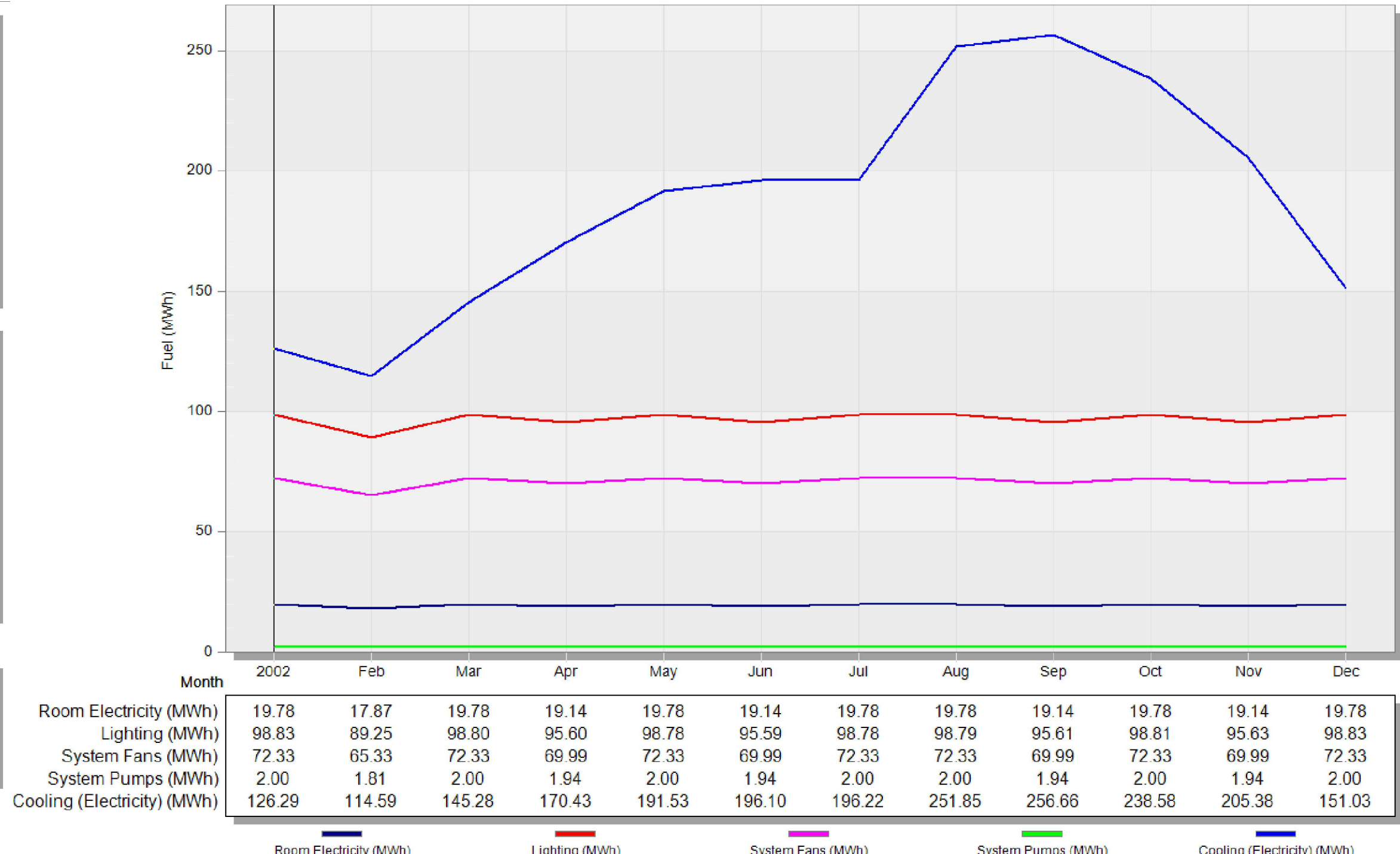
CO₂ Emissions



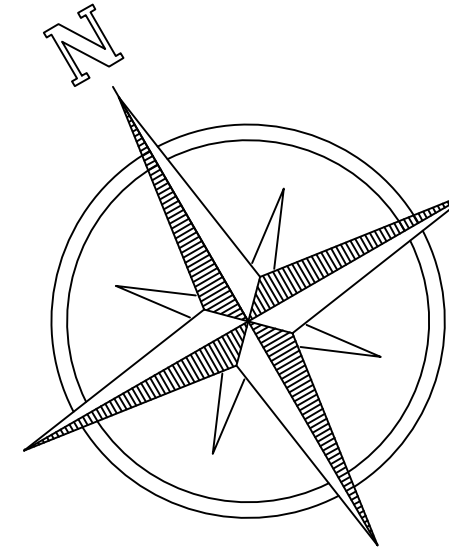
Comfort



Fuel Breakdown



COLLEGE OF ARCH. & PLAN
DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER

Design Studio (10), Technical
Project, 2025/2026

INSTRUCTORS:

Prof. Othman Alshamrani (Co)
Prof. Mohammed Essam Shaawat
Dr. M. Abdul Mujeebu
Dr. Mohammad Farooqi
Dr. Saiful Islam
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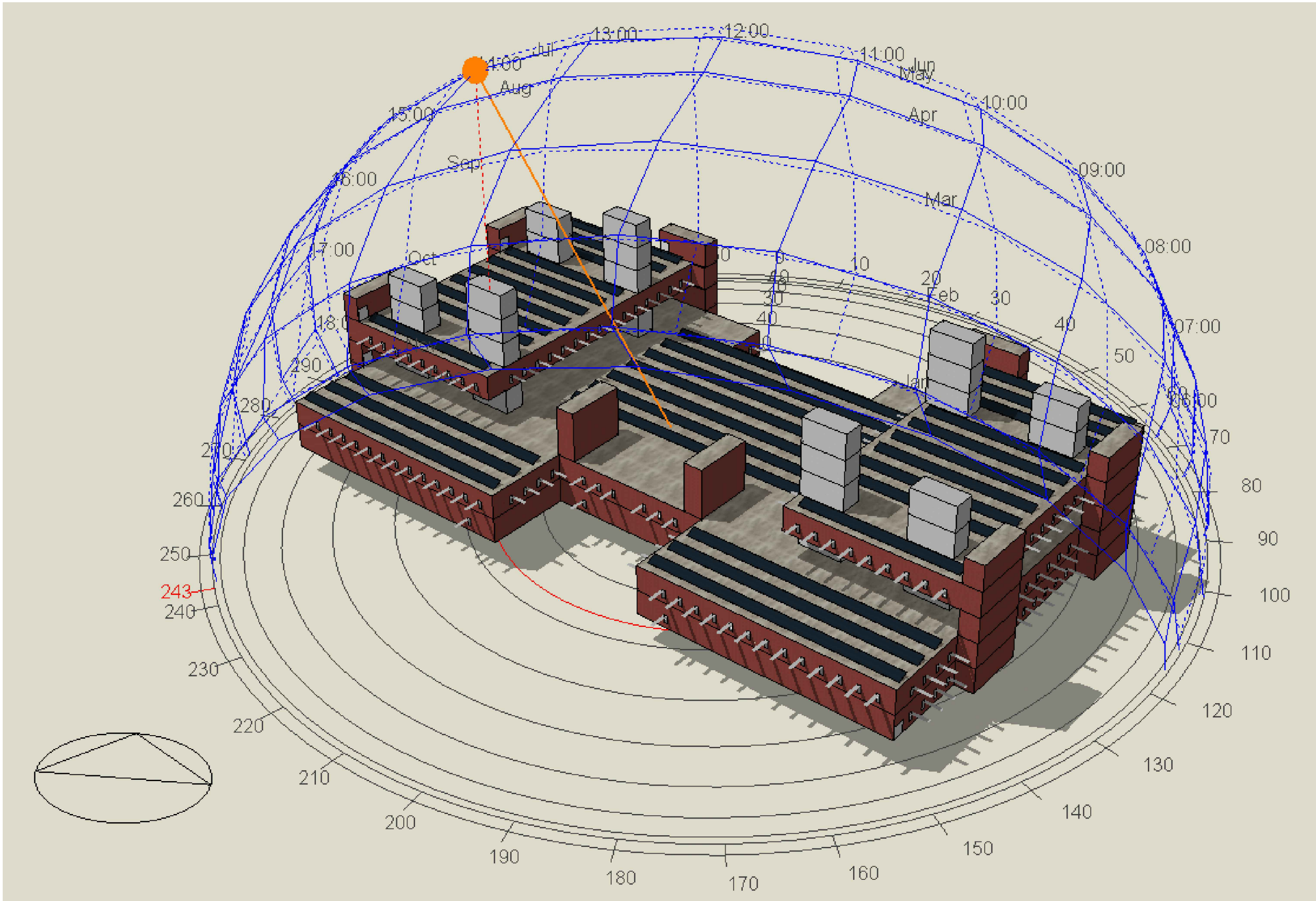
DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Emergency Hospital

DRAWING NAME:
Results

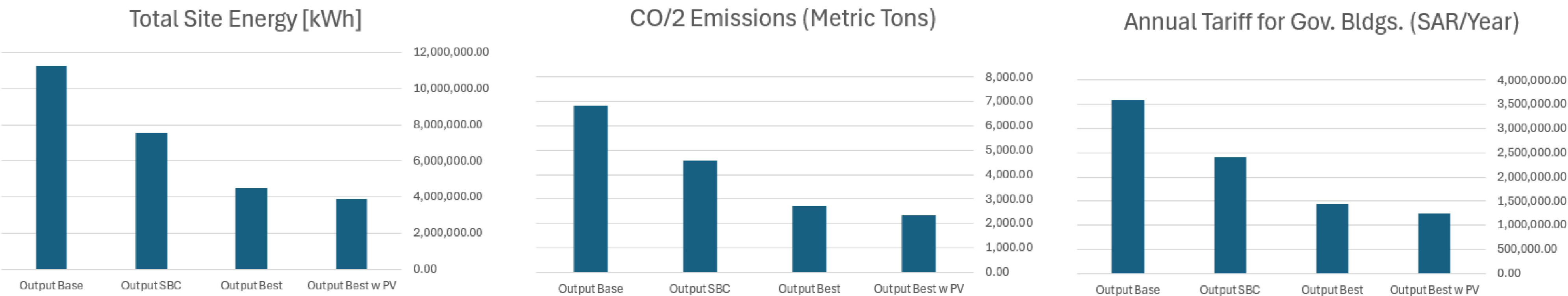
SCALE:
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SHEET NUMBER:
A63



Results of Best Case w PV

Simulation Parameters	Output Base	Output SBC	Output Best	Output Best w PV
Total Site Energy [kWh]	11,251,564.23	7,548,217.84 (-32.91%)	4,515,270.26 (-59.87%)	3,866,985.77 (-65.63%)
Energy Per Total Building Area (kWh/m ²)	634.80	425.86 (-32.91%)	254.75 (-59.87%)	218.17 (-65.63%)
Annual Tariff for Gov. Bldgs. (SAR/Year)	3,600,500.55	2,415,429.71 (-1,185,070.84)	1,444,886.48 (-2,155,614.07)	1,237,435.45 (-2,363,065.1)
CO ₂ Emissions (Metric Tons)	6,820	4,570 (-32.99%)	2,740 (-59.82%)	2,340 (-65.69%)



PV CHARACTERISTICS	Input
PANEL NAME	N-Type ABC
SOLAR CELLS	Monocrystalline
NO. OF CELLS	108(6*18)
NO. OF PANELS	816
EFFICIENCY MAX (%)	23.6
P MAX (W)	470
DIMENSIONS (mm)	1757x1134x30

Simulation Parameters	Output
PHOTOVOLTAIC POWER GENERATED (kWh)	682037.449
ENERGY SAVING (%)	15.11

Summary for Design and Construction Rating Systems

Note: All information on this tab is READ-ONLY. To edit, see previous tabs.

EA Credit Renewable Energy

Tier 1: On-site renewable energy percent of annual project site energy use	15.1%
Tier 2: New Off-site renewable electricity percent of annual project site energy use	0.0%
Tier 3: Off-site renewable percent of annual project site energy use	0.0%
Points:	4 Points

Summary

Note: All information in this section is READ-ONLY. To edit, see previous tabs.

Total energy cost savings (excluding site-generated renewable energy) (%) 65.7%

For projects pursuing EA Credit Optimize Energy Performance

Total energy cost savings (including site-generated renewable energy) (%) 59.9%

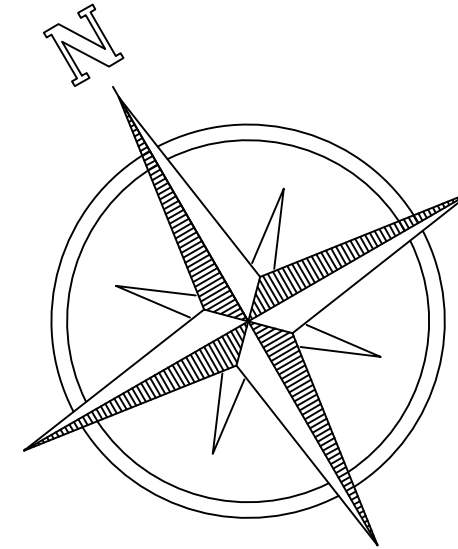
Performance Threshold

Thresholds are based on the major renovation area as a percentage of total conditioned building area.

Performance Improvement Threshold																								
5%	6%	8%	10%	12%	14%	16%	18%	20%	22%	24%	26%	29%	32%	35%	38%	42%	46%	50%						54%
Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18						EP



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DEPT. OF BUILDING ENGINEERING



ALL UNITS IN METER

Design Studio (10), Technical
Project, 2025/2026

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Dr. Fahad Alyami
Engr. Meqdad Hasan
Engr. Rehan Jamil

DONE BY:
ABDULELAH ALQABBANI

PROJECT:
Emergency Hospital

DRAWING NAME:
Solar Panels Case

SCALE:
--

SHEET NUMBER:
A64